

# 3D-Experience



## 3D Experience

### Arena Design System (ADS)

Version: 1.0

Status: Draft

Technologies

- Three.js
- React Three Fiber
- Drei
- GSAP
- Framer Motion
- Blender
- Spline

## Overview

ArenaMind AI is designed as a **Spatial AI Operating System**.

Instead of navigating through flat dashboards, users interact with a living, breathing representation of the stadium.

The stadium itself becomes the primary interface.

Every operational insight is visualized spatially.

Examples include:

- Crowd density
- Security incidents
- Transportation

- Parking
  - Volunteer locations
  - Emergency routes
  - Energy consumption
  - Weather
  - Accessibility
- 

## Design Vision

| “Imagine walking into the FIFA World Cup Mission Control.”

Users should immediately feel immersed inside an intelligent digital environment.

The interface should combine:



Stadium



World Map



Artificial Intelligence



Real-time Data



Interactive 3D

into a single cohesive experience.

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## Experience Goals

The 3D layer should help users answer:

- Where is the problem?
- What is happening?
- Why is it happening?
- What should happen next?

Rather than reading tables,  
users explore the stadium naturally.

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# Design Principles

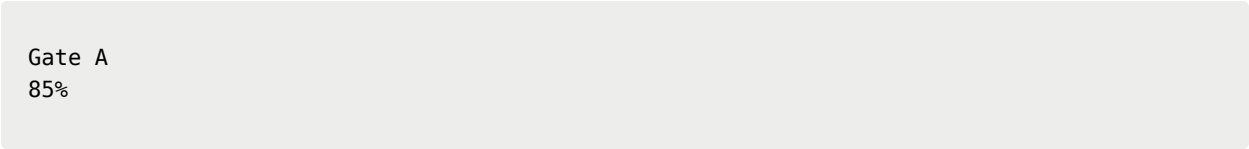
## 1. Spatial Computing

Information should exist within space.

Not inside floating spreadsheets.

Example

Instead of



Gate A  
85%

Show

A glowing Gate A within the stadium  
with animated crowd particles.

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## 2. Living Stadium

The stadium never stops moving.

Ambient motion includes

- crowd movement
- stadium lighting
- AI holograms
- drone routes
- transport movement
- weather
- particles

The application should never appear frozen.

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### 3. Direct Manipulation

Users interact directly with objects.

Rotate

Zoom

Pan

Select

Inspect

Simulate

Instead of opening multiple pages.

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### 4. Information on Demand

Show only essential information.

Additional information appears when users

Hover

Click

Search

Ask AI

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## 3D Scene Architecture

Scene

|

├─ Sky Dome

├─ Stadium Environment

├─ Stadium Model

- └ Crowd Simulation
- └ AI Holograms
- └ Weather Layer
- └ Lighting
- └ Transport Layer
- └ Drone Layer
- └ Digital Twin
- └ Emergency Overlay
- └ Particle Engine
- └ Camera

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## Landing Experience

The first impression should be unforgettable.

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### Intro Animation

Camera begins above Earth.

↓

Zooms into host city.

↓

Clouds separate.

↓

Stadium appears.

↓

Lights turn on.

↓

Digital Twin activates.



AI Nodes connect.



Navigation appears.

Total duration

8–10 seconds.

Users can skip.

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## Hero Scene

The hero section contains

- Stadium
- Floating holograms
- AI data streams
- Particle systems
- Rotating Digital Twin
- Dynamic lighting
- Moving camera

Everything remains interactive.

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## Stadium Model

The stadium is the primary interaction surface.

Supported interactions

- Rotate
- Pan
- Zoom

- Reset
  - Focus on zone
  - AI highlight
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## Interactive Zones

Stadium

└─ Parking

└─ Entrances

└─ Security

└─ Seating

└─ VIP

└─ Medical

└─ Food Court

└─ Restrooms

└─ Merchandise

└─ Media

└─ Volunteer Base

└─ Emergency Exits

Every zone is selectable.

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## Hover Behavior

Hovering over a zone

↓

Highlights geometry.



Displays label.



Shows metrics.



Displays AI summary.

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### Example

Gate B

Capacity

82%

Crowd

High

Prediction

Congestion in 8 min

Confidence

96%

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## Click Behavior

Clicking a zone opens

Glass Panel

containing

Live metrics

Crowd history

Predictions



Incidents

AI actions

Camera feeds

Transport status

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## Camera System

ArenaMind AI uses cinematic camera movement.

Never instant jumps.

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## Camera Modes

Overview

Entire stadium.

Inspection

Focus on one zone.

Emergency

Zoom to incident.

AI Focus

Moves toward prediction.

Presentation

Auto fly-through.

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## Camera Animation

Transitions

- Smooth
- Physics based
- Eased

- Continuous

Never abrupt.

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## Digital Twin

The Digital Twin is the heart of ArenaMind AI.

It continuously mirrors

- Crowd
- Infrastructure
- Weather
- Security
- Transport
- Utilities

in real time.

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## Layers

Digital Twin

└ Crowd

└ Transport

└ Weather

└ Medical

└ Security

└ Energy

└ Accessibility

└ Parking

└ Utilities

Users can toggle layers independently.

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## Crowd Visualization

Crowds are represented using animated particles.

Particle color

Green



Yellow



Orange



Red

Particle density represents occupancy.

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## Crowd Flow

Particles move continuously.

AI predictions display

future movement

using animated projected paths.

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## AI Holograms

Every important metric appears as floating holograms.

Examples

Gate A

82%



Crowd



Prediction



Recommendation

Panels always face the camera.

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## Emergency Visualization

Incidents immediately affect the stadium.

Example

Fire



Red pulse



Camera zoom



Nearby routes glow



Emergency exits illuminate



Volunteer locations appear



AI recommendation panel opens

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## Transportation Layer

Animated

Metro

Bus

Parking

Ride Sharing

Walking

Routes

Vehicles move in real time.

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## Weather Layer

Weather influences

- Crowd movement
- Visibility
- Lighting
- Route suggestions

Rain

↓

Cloud animation

↓

Wet reflections

↓

Umbrella recommendations

↓

## Lighting System

Lighting communicates system health.

Green

Healthy

Blue

Operational

Yellow

Attention

Orange

Warning

Red

Emergency

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## Drone Layer

Visualize

Security drones

Medical drones

Broadcast drones

Animated flight paths.

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## Accessibility Layer

Displays

Accessible entrances

Wheelchair routes

Elevators

Hearing assistance

Audio guidance

All highlighted spatially.

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## Sustainability Layer

Visualize

Energy

Water

Carbon

Waste

Lighting

HVAC

as glowing overlays.

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## Particle Systems

Particle systems communicate

AI

Weather

Crowds

Transport

Neural networks

Energy

Data flow

Particles should remain subtle.

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# AI Visualization

Artificial Intelligence should feel alive.

Examples

Neural connections

Gradient pulses

Flowing energy

Reasoning waves

Prediction halos

Confidence glow

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# Map Integration

Google Maps transitions into

3D Stadium

without leaving the page.

Zoom levels

World

↓

Host Country

↓

Host City

↓

Stadium

↓

Zone

↓



Seat

Smooth transitions.

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## User Interaction

Supported

Desktop

- Mouse
- Keyboard

Tablet

- Touch
- Pinch

Mobile

- Touch
  - Gyroscope (optional)
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## Motion Behavior

Hover

Glow

↓

Scale

↓

Elevation

Selection

Camera

↓

Zoom



Glass Panel



AI Summary

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## Performance Budget

Target

Desktop

120 FPS

Minimum

60 FPS

Scene loading

<2 seconds

Interaction latency

<100 ms

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## Accessibility

Provide alternative experiences.

Reduced Motion



Static stadium



No particles



Instant camera



No parallax

Support

prefers-reduced-motion

## Technical Stack

| Feature           | Technology        |
|-------------------|-------------------|
| 3D Rendering      | Three.js          |
| React Integration | React Three Fiber |
| Helpers           | Drei              |
| Animations        | GSAP              |
| UI Motion         | Framer Motion     |
| Models            | Blender           |
| Materials         | Three.js PBR      |
| Environment       | HDRI              |
| Performance       | Instancing + LOD  |

## Future Enhancements

- AR stadium navigation
- VR command center
- Apple Vision Pro support
- Multi-user collaboration
- AI holographic assistant
- Live drone telemetry
- Real-time seat occupancy

- Gesture interactions
  - Spatial voice commands
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## Experience Checklist

Every 3D screen must provide:

- Interactive camera
  - Smooth transitions
  - Spatial navigation
  - Live operational overlays
  - AI holograms
  - Crowd visualization
  - Layer controls
  - Responsive performance
  - Accessible fallback
  - Consistent visual language
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## Inspiration

The ArenaMind AI 3D experience draws inspiration from:

- Apple Vision Pro
- Google Earth
- FIFA World Cup Broadcast Graphics
- Unreal Engine Stadium Visualizations
- Formula 1 Race Control
- Iron Man JARVIS Interface
- SpaceX Mission Control

- Arc Browser
  - Lusion Interactive
  - Bruno Simon Portfolio
  - Active Theory
  - Three.js Journey
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## ArenaMind Spatial Experience

**“Don’t build dashboards. Build worlds.”**

ArenaMind AI transforms the stadium from a static venue into a living digital ecosystem where AI, data, and human decisions converge in real time.